

Neural Population Dynamics from Internal Representations in Learned Obstacle Avoidance

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Introduction

Artificial intelligence controllers remain difficult to interpret despite their success in autonomous navigation.

We propose a neuroscience-inspired framework that analyses hidden-layer activity as neural population recordings.

The objective is to understand how internal representations support adaptive obstacle avoidance under external perturbations.

Key Contributions

- Neuro-inspired analysis of PPO hidden representations
- Low-dimensional neural manifolds
- Behavioural robustness under perturbations
- Distributed neural encoding
- Explainable reinforcement learning

Methods

- PPO controller
- Continuous obstacle avoidance
- External perturbations
- PCA
- Neural decoding
- Hidden-unit tuning

figure1.png

figure2.png

Key Findings

- PPO learned structured low-dimensional neural representations.
- Behaviour changed systematically with perturbation magnitude.
- Hidden units exhibited distributed tuning properties.

- Neural population trajectories evolved smoothly during successful goal-directed behaviour.

- Internal representations remained interpretable using standard neuroscience analysis techniques.

Future Work
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Results

The trained PPO controller exhibited robust adaptive behaviour under multiple perturbation conditions. Neural analyses revealed structured latent representations that evolved continuously during obstacle avoidance.

Behavioural Performance

Behavioural metrics demonstrated significant effects of perturbation magnitude on reward, duration, path length, heading deviation and lateral error (Kruskal-Wallis, $p < 0.001$). Maximum speed remained invariant.

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Figure 3. Behavioural performance across perturbation conditions.

Neural Population Dynamics

Principal Component Analysis showed that hidden-layer activity evolved along a structured low-dimensional manifold. Neural trajectories remained smooth throughout successful goal-directed movement.

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Figure 4. Low-dimensional neural manifold revealed by PCA.

Hidden Unit Tuning

Representative hidden units exhibited diverse tuning relationships with movement speed, indicating distributed neural encoding rather than single-unit specialization.

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Figure 5. Hidden-unit tuning curves.

Conclusion

Hidden-layer activity contained meaningful neural population

Interpretability

Neural population analyses revealed that internal representations encode behaviourally meaningful information that remains linearly accessible throughout movement.

These findings support a neuroscience-inspired framework for interpreting reinforcement learning controllers.

Neural Decoding

Linear decoders successfully recovered task-relevant variables from hidden-layer activity, demonstrating that latent representations preserve behaviourally meaningful information.

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Figure 6. Linear decoding of task variables.

Decision Making

The controller selected alternative trajectories according to the perturbation condition, demonstrating adaptive route selection instead of stereotyped behaviour.

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Figure 7. Route selection and perturbation statistics.

Robustness

Behaviour remained stable across most perturbation conditions. Only the strongest perturbations produced substantial degradation in performance, indicating graceful failure rather than catastrophic collapse.

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Figure 8. Robustness across evaluation episodes.

Discussion

Rather than acting as an opaque controller, the trained PPO policy

